

# Practice Problem Set 1

ECON 204 – Contemporary Economic Issues

Winter 2026

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## Part I: Inflation, Cost of Living, and Monetary Policy

### Question 1: Measuring Inflation

The Consumer Price Index (CPI) was 152.0 last year and 158.1 this year.

1. Calculate the inflation rate.
2. If a worker's nominal wage increased by 3.5%, did their real wage rise or fall?
3. Explain what your result implies about purchasing power.

### Question 2: Cost of Living and Household Budgets

A household spends:

- 30% on housing
- 20% on food
- 15% on transportation
- 10% on utilities
- 25% on other goods

During the year:

- housing costs rise 8%
- food prices rise 5%
- transportation rises 4%
- utilities rise 6%
- other goods rise 2%

1. Calculate the household-specific inflation rate.
2. Explain why this household's inflation rate may differ from the national CPI.
3. Which households are most affected when housing costs rise faster than other prices?

### Question 3: Real Interest Rates and Monetary Policy

The nominal interest rate is 5.5% and expected inflation is 2.8%.

1. Calculate the real interest rate.
2. Explain how the real interest rate influences consumption and borrowing decisions.
3. If the Bank of Canada raises the policy rate, explain the channels through which inflation may decline.

### Question 4: Inflation Expectations

Suppose workers expect inflation to rise from 2% to 5%.

1. Explain how wage bargaining may change.
2. Explain how firms may respond.
3. Describe how expectations can create persistent inflation.

## Part II: Income Inequality

### Question 5: Measuring Inequality

Five households earn the following annual incomes:

| Household | Income (\$) |
|-----------|-------------|
| A         | 20,000      |
| B         | 30,000      |
| C         | 45,000      |
| D         | 70,000      |
| E         | 135,000     |

1. Calculate the mean income.
2. Calculate the median income.
3. Explain why the mean may differ from the median.

### **Question 6: Income Shares**

Using the data above:

1. Calculate total income.
2. Calculate the income share of the top household.
3. Calculate the income share of the bottom two households.
4. Interpret what these shares imply about inequality.

### **Question 7: Policy and Redistribution**

Suppose the government introduces a transfer of \$8,000 to each of the two lowest-income households, financed by a \$16,000 tax on the highest-income household.

1. Calculate new incomes.
2. Calculate the new mean income.
3. Has inequality increased or decreased? Explain.

### **Question 8: Poverty Measurement**

Assume the poverty threshold is \$28,000.

1. Which households are below the poverty line?
2. Calculate the poverty rate.
3. Explain one limitation of using a poverty line to measure economic hardship.

## **Part III: Housing Affordability and Real Estate Markets**

### **Question 9: Price-to-Income Ratio**

A household earns \$95,000 annually. The average home price is \$760,000.

1. Calculate the price-to-income ratio.
2. Interpret what this ratio implies about affordability.
3. Explain why this measure may not fully capture affordability.

### **Question 10: Mortgage Payments and Interest Rates**

A household borrows \$500,000 with a 25-year mortgage.

1. Estimate how monthly payments change if the mortgage rate rises from 2.5% to 5.5%.
2. Explain why rising interest rates can worsen affordability even if house prices stabilize.

### **Question 11: Rental Affordability**

A renter earns \$3,800 per month. Rent increases from \$1,200 to \$1,850.

1. Calculate the rent-to-income ratio before and after.
2. At what threshold is housing considered unaffordable?
3. Explain how rising rents contribute to inequality.

### **Question 12: Supply Elasticity and Prices**

Explain how each of the following affects housing affordability:

1. restrictive zoning regulations
2. increased immigration
3. new transit infrastructure
4. a decline in mortgage interest rates

## Part IV: Integrated Policy Analysis

### Question 13: Inflation and Housing

Explain how high inflation and rising interest rates can simultaneously:

- slow house price growth
- increase mortgage payments
- raise rental demand

### Question 14: Inequality and Housing Wealth

Explain how rising house prices can increase wealth inequality between:

- homeowners and renters
- older and younger households

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*End of Practice Set*